DERIVED ADIC COMPLETION AGREES WITH USUAL COMPLETION FOR FINITE MODULES TENSORED WITH A FLAT MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted proof span contains the author's sentence 'By Lemma ... we conclude R lim (M tensor_A N) tensor^L_A A/I^n only has cohomology in degree 0 given by the usual completion ...', whose slightly elliptical wording is reproduced verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The system Tor^A_i(M, A/I^n) is pro-zero for i > 0 when M is finite over a Noetherian ring. invoked at proof lines 27333-27335. Is the input whose flat base change yields the vanishing needed for (1).
- The computation of the derived limit of the base changed Koszul-type system, giving the usual completion in degree 0. invoked at proof lines 27339-27342. Identifies the derived completion with the usual completion.
- Flatness of a localization A -> A_f and the identification M_f = M tensor_A A_f. used at proof line 27343. Derives part (2) from part (1).

Source reference. Original lemma 27245: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27245–27245. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 6450: sources/stacks-more-algebra/pinned-original-source.tex, source lines 6450–6450. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let I be an ideal in a Noetherian ring A.*

(1) *If M is a finite A-module and N is a flat A-module, then the derived I-adic completion of $M \otimes_A N$ is the usual I-adic completion of $M \otimes_A N$.*

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(2) *If M is a finite A-module and $f \in A$, then the derived I-adic completion of M_f is the usual I-adic completion of M_f.*

Proof. For an A-module M denote $M^\wedge$ the derived completion and $\lim M/I^nM$ the usual completion. Assume M is finite. The system $\mathrm{Tor}_i^A(M, A/I^n)$ is pro-zero for $i > 0$, see Lemma 6450. Since $\mathrm{Tor}_i^A(M \otimes_A N, A/I^n) = \mathrm{Tor}_i^A(M, A/I^n) \otimes_A N$ as N is flat, the same is true for the system $\mathrm{Tor}_i^A(M \otimes_A N, A/I^n)$. By Lemma 27245 we conclude $R\lim(M \otimes_A N) \otimes_A^{\mathbf{L}} A/I^n$ only has cohomology in degree 0 given by the usual completion $\lim M \otimes_A N/I^n(M \otimes_A N)$. This proves (1). Part (2) follows from (1) and the fact that $M_f = M \otimes_A A_f$. □